

SMART POLYPHARMACY & VITAMIN RECOMMENDATION SYSTEM FOR GERIATRIC PATIENTS

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POLYPHARMACY RISK SCORE SYSTEM BASED ON DRUG INTERACTION

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

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ABSTRACT

Polypharmacy is an emerging issue in contemporary healthcare, particularly in geriatric patients who are likely to take multiple medications to manage chronic illnesses. Even though the combination of several medications can be clinically justified, it predisposes the risk of drug-drug interactions, adverse drug reactions, complications in treatment, and preventable hospitalizations. Current methods of polypharmacy evaluation usually consider more general aspects like the number of drugs, age, and comorbidities. Nevertheless, a lot of these methods fail to incorporate the severity of drug-drug interactions into a patient-specific risk score. This poses a disconnect in clinical decision support since two patients with an identical number of medicines can have vastly different amounts of risk based on the severity of interaction in their medication profile.

The study introduces a Polypharmacy Risk Score System Based on Drug Interaction that was created as a component of the Smart Polypharmacy and Vitamin Recommendation System in Geriatric Patients. The primary objective of this personal study is to determine a personalized polypharmacy risk score and the risk level according to the severity of drug interactions and the choice of patient-specific clinical factors. The adopted system applies machine learning to the drug interaction-related prediction aspect, and other clinical variables, including age, liver function, kidney function, existing diseases, and gender are directly entered into the user interface. The resulting risk score is determined through an expert-informed weighted formula, which is a combination of medication burden, patient age, the severity of drug-drug interaction, liver function, and kidney function. This formula was formulated based on expert knowledge and tested by Dr. Rohan Jayasooriya.

The solution implemented is a Next.js frontend, a Flask-based backend API, drug and disease search functionality, a drug interaction knowledge base, a drug interaction model based on a random forest, an adverse drug event prediction model, and a dashboard to display the final risk score, risk level, interaction summary, and predicted adverse drug events. Besides the primary risk scoring purpose, the study adds an extension of the system to predict potential adverse drug events based on drug, age, gender, existing disease, and predicted ADE-related information. The adverse drug event model predicts probable ADE outcomes with the help of encoded categorical features and a Random Forest Classifier.

The last system is an interpretable and organized clinical decision-support system that finds high-risk polypharmacy cases. It helps find dangerous drug interactions early on and gives the results in a way that patients, caregivers, and health professionals can understand. The study adds a practical, implementation-based model of integrating drug interaction intelligence, expert-based scoring, and adverse event prediction in geriatric medication safety.

Keywords: Polypharmacy, Drug-Drug Interaction, Risk Score, Adverse Drug Event, Random Forest, Clinical Decision Support, Geriatric Patients.

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I am especially thankful that Dr.Rohan Jayasooriya confirmed the expert-informed polypharmacy risk scoring formula that was employed in this study. His clinical feedback served to reinforce the practical applicability of the scoring logic and enhance the interpretability of the resulting risk assessment framework.

I would also like to recognize the researchers and authors whose published works contributed to developing the literature review and defining the gap in research in polypharmacy risk assessment, the severity of drug-drug interaction, clinical decision support, and adverse drug event prediction. Their work offered the scientific basis to the creation of the system that is not limited to counting the medication but takes the severity of the interaction as a significant element of patient-specific risk scoring.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ADE	Adverse Drug Event
ADR	Adverse Drug Reaction
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma Separated Values
DDI	Drug-Drug Interaction
EB-PRS	Evidence-Based Polypharmacy Risk Score
EHR	Electronic Health Record
EMR	Electronic Medical Record
eGFR	Estimated Glomerular Filtration Rate
JSON	JavaScript Object Notation
ML	Machine Learning
MRCI	Medication Regimen Complexity Index
PRS	Polypharmacy Risk Score
REST	Representational State Transfer
STOPP/START	Screening Tool of Older People's Prescriptions / Screening Tool to Alert to Right Treatment

Table 1 List of abbreviations

1. INTRODUCTION

Polypharmacy is usually described as the use of two or more drugs by a patient. Five or more medications are regarded as a sensible cut-off point in the detection of polypharmacy in numerous clinical and research settings [1], [2]. This is especially prevalent in geriatric patients due to the fact that older adults tend to have a number of chronic diseases that need lifelong medication treatment. Despite the clinical need of using multiple medicines, it predisposes the development of adverse drug reactions, drug-drug interactions, decreased medication adherence, hospitalization, and poor patient outcomes [1], [3].

The risk of polypharmacy is more severe in the case of elderly patients due to the fact that ageing influences the absorption, distribution, metabolism, and excretion of medicines. Older patients may be susceptible to medication-related harm due to reduced liver function, reduced kidney function, frailty, and multimorbidity [3], [4]. Thus, polypharmacy cannot be evaluated based on the number of medicines. The actual risk of a patient is determined by the nature of medicines, the intensity of interactions between the medicines and the clinical status of the patient.

Drug-drug interaction (DDI) is one of the most significant clinical issues in polypharmacy. A DDI is the alteration in the effect, metabolism, concentration, or safety profile of a drug by another drug [5]. These interactions may decrease the effectiveness of treatment, enhance toxicity, or lead to adverse drug events. The more medicines, the more possible combinations of drugs, and the more likely it is that the interactions will be harmful [1]. This is particularly relevant in geriatrics whereby a patient can be prescribed by several healthcare professionals.

Current polypharmacy evaluation instruments and clinical decision-support systems are useful in identifying risks associated with medications. Nonetheless, a great number of them primarily rely on the general factors, including the number of medications, age, comorbidity, or potentially inappropriate medication lists [6], [7]. These methods are helpful, but they do not necessarily combine the severity of drug-drug interactions with a personalized risk score. Consequently, two patients with same number of medications can be given a similar risk level although one patient can have minor interactions whereas the other patient can have severe or life-threatening interactions.

In order to overcome this drawback, this study is aimed at the development of Polypharmacy Risk Score System Based on Drug Interaction as a part of the larger project named Smart Polypharmacy and Vitamin Recommendation System to Geriatric Patients. The primary aim of the given research is to determine a patient-specific polypharmacy

risk score and the risk level that relies primarily on the severity of drug interactions, but also includes some patient-specific variables, including the number of medications, age, liver functioning, and kidney functioning. Another negative drug event prediction system is also a part of the system based on drug, age, gender, existing diseases, and predicted ADE-related data.

The suggested system integrates clinical decision-support web-based functionality, machine learning, and expert knowledge. The drug interaction-related prediction and adverse drug event prediction components are primarily predicted using the machine learning model, with the other clinical values being directly entered by the user via the system interface. The risk score is eventually obtained by an expert-guided formula that is tested by Dr. Rohan Jayasooriya. Thus, the system is not only developed to identify drug interactions but also to transform the results into a risk score and a risk level that can be interpreted and used to make safer decisions regarding medication.

1.1 Background and Literature Survey

Polypharmacy is now a significant issue in health care due to the growing number of older people with multiple chronic conditions. Delara et al. reviewed and meta-analysed the literature and found that polypharmacy is prevalent among adults, particularly older adults and inpatients [1]. They also found that polypharmacy is linked to adverse health outcomes including drug interactions, hospitalisation, medication non-adherence, falls and death [1]. This indicates polypharmacy is a medication safety issue as well as a medication management issue.

While there is no one definition for polypharmacy, the most common definition used in research is five or more medications [1], [2]. Masnoon et al. demonstrated that definitions of polypharmacy range from simple counts to more descriptive terms, such as minor, major or excessive polypharmacy [2]. The variability of definitions poses a challenge for comparing research evidence and for implementing a single method of risk assessment in practice.

Polypharmacy is not always inappropriate. Polypharmacy is sometimes appropriate, when the use of multiple medicines is required for multiple chronic conditions. But inappropriate polypharmacy is when the harms of the medication regimen outweigh the benefits, when unnecessary medications are continued, or when adverse drug interactions

are not recognised early [3], [6]. Halli-Tierney et al. noted that medication review and deprescribing are essential for minimising the burden of unnecessary medications in the elderly [6].

Drug-drug interaction is a significant factor contributing to polypharmacy-related harms. When patients are taking multiple drugs, one drug can affect the absorption, metabolism, efficacy or toxicity of the other drug [5], [8]. Quinney identified that big data and computational science offer opportunities to better identify DDI risk, but also the problems of using big data in clinical practice [8]. Likewise, computerised drug interaction checking systems can help ensure drug safety, but are often used in isolation rather than as part of a comprehensive risk scoring system [9].

Electronic systems have been developed to assist with polypharmacy in the elderly. Mouazer et al. conducted a review of electronic systems for the management of polypharmacy and showed that current systems vary in design, knowledge sources, outputs and evaluation [7]. They found that most systems are rule-based or documentary systems and that few systems completely integrate multiple decision-support techniques. This suggests that a more comprehensive system is needed that can incorporate drug interaction knowledge, patient characteristics and predictive methods.

Current methods such as STOPP/START criteria and the Medication Regimen Complexity Index are helpful in identifying inappropriate medications and complex medication regimens [6], [7]. However, these approaches may be cumbersome to use in clinical practice as they require detailed clinical interpretation and may not automatically adapt to prescription changes. Also, many of these tools do not translate DDI severity to a final risk score for patients. So, an automated tool that can detect DDIs, aggregate severity, calculate a risk score and display it in a dashboard can be useful.

There is also growing interest in the application of machine learning in medication safety and polypharmacy risk modeling. Machine learning can find patterns in structured medication and patient information that may not be obvious to humans [10], [11]. But in

clinical decision support, the results of models should be understandable and relate to clinical reasoning. Therefore, the system in this study does not rely solely on machine learning. It uses machine learning to support prediction but also uses an expert-developed scoring formula to ensure that the final result is interpretable by users.

Literature shows the need. A real-world polypharmacy risk calculator should consider the absolute number of medications, age, organ function and in particular, the degree of drug interactions. It should also output a comprehensible risk and future adverse event awareness. This study provides such a system in the form of a web application that provides a personalized polypharmacy risk and adverse drug event risk.

1.2 Research Gap

While several studies have highlighted the risks associated with polypharmacy, there remains a need for a standardized, dynamic and individualized risk scoring system that incorporates the severity of drug interactions. Current models can rely on the number of medications, age, and illness burden to calculate risk [1], [2], [6]. While these variables are important, they do not fully capture the risk of a patient's medication regimen.

The main problem with current systems is that drug interactions are often considered simply as a yes or no factor, for example, the presence or absence of an interaction. This does not account for the varying severity of interactions. For instance, a low-severity interaction may only warrant monitoring, whereas a high-severity interaction may warrant medication review and/or clinical action. So, a risk scoring model that ignores interaction severity may underestimate or overestimate a patient's risk.

A further shortcoming is that many current polypharmacy assessment tools are static. A static assessment may not be responsive to a change in a patient's medication, the addition of a new medication, or a change in the patient's health (such as liver or kidney function).

This is a major gap because the risk of polypharmacy is dynamic; it may change as the patient's medication use and clinical conditions change [7], [10].

There is also a need for the integration of polypharmacy risk assessment with adverse drug event prediction. Current tools may predict a high risk of medication use but they may not provide an additional prediction of what type of adverse drug event may occur. Given that one of the major consequences of polypharmacy is adverse drug events, including an ADE predictor can enhance the usefulness of a decision-support tool [11].

In addition, many models and tools are developed and/or validated primarily in Western healthcare systems. This could limit their value in the Asian and low-middle-income healthcare settings, where drug prescribing patterns, drug availability, patient demographics, and health care processes may be different [10], [12]. Hence, an implementation-based, adaptable system is required.

The issues have identified the research gap that this study addresses as:

There is a lack of an implementation-based, patient-specific polypharmacy risk score system that integrates drug–drug interaction severity with clinical risk factors and extends the assessment by predicting possible adverse drug events

Finally, many systems do not provide a practical academic-to-implementation bridge. They may present a theoretical score or a standalone model but lack a usable interface, validation workflow, API routes, dashboard, and report export. This project fills that implementation gap by delivering a working full-stack component that can be integrated into the wider geriatric care platform.

Feature / Capability	Existing drug-count tools	Interaction checkers	Machine learning studies	Proposed system
Medication count included	Yes	No / limited	Varies	Yes
Patient age included	Often	No / limited	Often	Yes
DDI severity integrated into score	Rare	Listed separately	Varies	Yes
Liver and kidney function included	Rare	No	Varies	Yes
ADE prediction included	No	No	Yes	Yes
Dynamic web assessment	Limited	Sometimes	Research prototype	Yes
Report export for clinician review	Rare	Limited	Rare	Yes

Table 2 Comparison of existing approaches with the proposed system

1.3 Research Problem

Risk assessment for polypharmacy is complex because health-care professionals need to consider multiple aspects. These are the number of medicines, the patient's age, medical conditions, organ function and the level of drug interactions. This assessment can be difficult to perform when done by hand, as clinically significant interactions may be overlooked, particularly when patients take many medicines.

Current systems are helpful, but they may not reflect the true risk level of a patient because they are based on the number of medicines or general clinical information [6], [7]. A patient who uses seven medicines may have a different risk level to another patient who uses seven medicines, depending on the level of risk of the drug interactions in their medicines list. So a more specific risk calculation is needed.

The research problem of this study is:

How does a software clinical decision-support system determine a patient-specific polypharmacy risk score and risk level by incorporating the severity of drug–drug interactions with patient characteristics, and predict potential adverse drug events by using machine learning?

There are three key requirements of this research problem. First, it should detect drug–drug interactions in the patient's medication list. Second, it should translate the severity of the drug–drug interaction and clinical factors into a risk score and risk level. Third, it should make an additional prediction of the risk of adverse drug events based on patient and medication information.

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this research is to design and develop a Polypharmacy Risk Score System Based on Drug Interaction that calculates a personalized risk score and assigns a risk level by integrating drug–drug interaction severity with selected patient-specific clinical factors.

The system seeks to address the limitations of current polypharmacy measures by taking into account not only number of medications and age, but also the severity of drug interactions, liver and kidney function. The outcome of the system is to assist medication safety management by providing low, moderate, high or very high polypharmacy risk levels.

1.4.2 Specific Objectives

1. To develop a drug interaction-based risk assessment system that accepts a patient's medication list and identifies possible drug–drug interactions.
2. To classify and summarize interaction severity so that the system can distinguish between different levels of medication risk.
3. To design and implement an expert-informed polypharmacy risk score formula using medication count, patient age, DDI severity, liver function, and kidney function.
4. To verify the scoring logic using expert knowledge, with clinical validation support from Dr. Rohan Jayasooriya.
5. To implement a machine learning-supported adverse drug event prediction model using drug, age, gender, and existing disease inputs.
6. To develop a user-friendly web interface that allows users to enter patient details, medication details, liver function, kidney function, and existing diseases.
7. To generate a dashboard output that displays the final risk score, risk level, drug interaction count, severity summary, adverse drug event predictions, and detailed risk calculation.
8. To evaluate the implemented system through functional testing, input validation, output verification, and consistency checking.

1.4.3 Scope of Research

The scope of this individual research is limited to the polypharmacy risk-scoring component. It does not prescribe medicines, stop medicines, or replace the judgement of a licensed healthcare professional. The system provides decision-support information using available patient inputs and knowledge-base/model outputs. The wider project includes other components such as vitamin deficiency detection, lifestyle support, and meal recommendation, but those functions are outside the direct evaluation scope of this report.

1.5 Significance of the Research

The significance of this research lies in its ability to convert medication safety knowledge into an implementable software workflow. In many healthcare environments, medication review depends heavily on the experience and available time of the clinician. A structured system can reduce the chance that serious interactions are overlooked during a busy consultation. The system does not remove the need for professional review; rather, it helps organise the information that should be reviewed.

The system is significant for geriatric patients because older adults are more likely to have multiple chronic illnesses and to receive long-term medicines. A patient may take medicines for blood pressure, diabetes, cholesterol, pain, sleep, gastric protection, and infection at the same time. When these medicines are checked one by one, the clinician may miss the combined effect of the whole regimen. A patient-level risk score helps summarise this combined burden.

The research is also significant because it treats interaction severity as a central feature. A medicine count can identify potential complexity, but it cannot explain clinical danger. By weighting major, moderate, and minor interactions differently, the system reflects the fact that not all interactions are equal. This improves the usefulness of the score and directly addresses the research gap identified in the proposal.

For healthcare providers, the system can be used as a screening tool before a detailed medication review. Patients with high or very high scores can be prioritised for pharmacist or doctor attention. Patients with low scores can still be monitored routinely. This prioritisation is valuable in environments where time and specialist resources are limited.

For caregivers, the system provides a more understandable view of medication risk. Caregivers may not understand technical pharmacology, but they can understand a dashboard that shows a risk category, interaction count, and report recommendation. This supports safer communication with healthcare professionals and may encourage patients to seek advice when needed.

For academic research, the system demonstrates how clinical decision-support logic, machine learning prediction, and full-stack software engineering can be combined. The project includes model loading, API design, frontend validation, cloud persistence, explainable score calculation, and exportable report generation. This makes the work suitable as an undergraduate final research project because it contains both research reasoning and practical implementation.

1.6 Report Organization

This report is organised into four main chapters. Chapter 1 introduces the background, literature, research gap, problem statement, objectives, scope, and significance of the study. It explains why polypharmacy risk assessment should include drug-drug interaction severity and why geriatric patients need personalised risk support.

Chapter 2 explains the methodology. It describes the system approach, data and knowledge-base integration, ADE prediction, EB-PRS formula, backend and frontend architecture, requirements achieved, commercialization aspects, ethical considerations, testing strategy, and tools used. This chapter connects the proposal idea with the actual implementation.

Chapter 3 presents the results and discussion. It explains the functional outcomes of the implemented system, demonstrates the risk score using controlled scenarios, discusses the research findings, and summarises the individual contribution. The chapter avoids fabricating clinical accuracy values and focuses on the verified implementation results available from the completed system.

Chapter 4 concludes the report by summarising the work, identifying limitations, and recommending future improvements. The reference list follows IEEE style, and the appendices provide additional implementation details, API structures, frontend types, validation logic, export content, and future enhancement plans.

2. METHODOLOGY

This chapter describes the methodology used to design and develop the Polypharmacy Risk Score System Based on Drug Interaction. The methodology aims to describe the process of designing, developing, and testing the proposed system. The system was developed as part of the individual research of the group project called Smart Polypharmacy & Vitamin Recommendation System for Geriatric Patients. This research is one of the four major sub-components of the group project, which aims to determine a patient-specific polypharmacy risk score based on the severity of drug–drug interactions and other clinical parameters.

The approach was developed considering clinical and technical needs. Clinically, the system must take into account the severity of drug–drug interactions, the number of medicines, age, liver function and renal function. From the technical side, it needed a robust data processing pipeline, machine learning capabilities, a backend engine to calculate risk, and a frontend dashboard. We chose this method because the risk of polypharmacy cannot be measured solely by the number of medications. Polypharmacy is known to be linked with various negative outcomes such as drug interactions, hospitalization, falls, non-adherence and death [1]. So, a more relevant system should be based on the number of medicines, severity of drug interactions and patient vulnerability.

The approach has five stages: requirement analysis and literature review, data collection and preprocessing, drug interaction knowledge-based integration, machine learning model development, and implementation of the expert-based polypharmacy risk score formula. The last stage is to integrate with the frontend dashboard and user interface, so that users can get the final risk score, risk level, interaction summary, and adverse drug event prediction.

System Overview and Diagrams

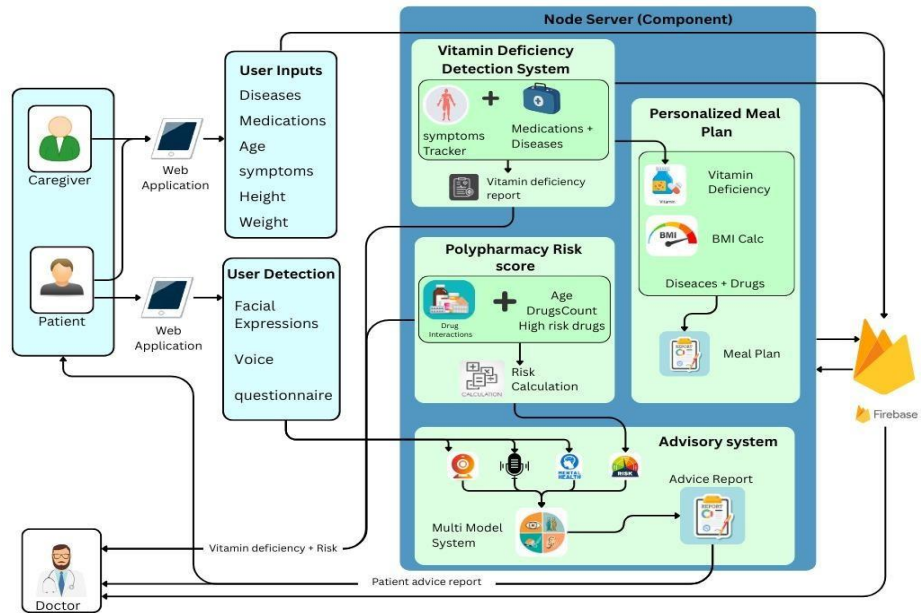


Figure 1 Overall System Diagram

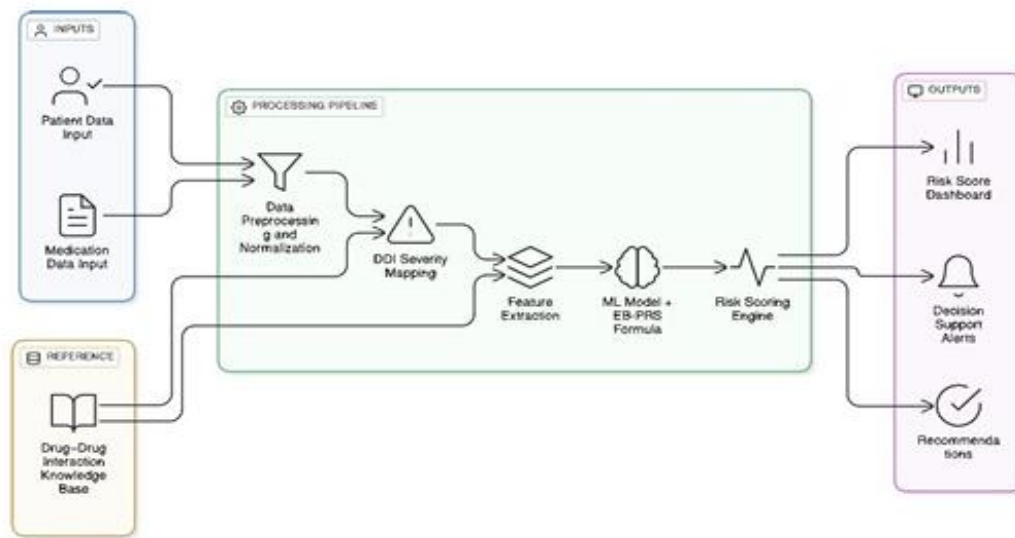


Figure 2 Individual System Diagram

2.1 Understanding Key pillars of the Research Domain

2.1.1 Polypharmacy

Polypharmacy is generally understood as the concurrent use of multiple medicines by a patient. Many studies commonly use the threshold of five or more medicines to identify polypharmacy [1], [2]. One, this definition is especially relevant in geriatric care because older adults often take several medicines for long-term chronic conditions such as diabetes, hypertension, heart disease, kidney disease, and neurological disorders.

However, the number of medicines alone does not fully explain patient risk. Some patients may use several medicines safely, while others may experience harmful interactions even with a smaller number of drugs. Therefore, this research treats medication count as one important factor, but not the only factor. The final system combines medication burden with drug interaction severity and patient-specific clinical details, such as age, health conditions, and other medications the patient may be taking.

2.1.2 Drug–Drug Interactions

Drug-drug interaction is a major risk factor in polypharmacy. DDI is defined as one drug affecting the effect, metabolism, absorption or toxicity of another drug [5]. The consequences of DDIs can be more severe in the elderly because of age-related changes in metabolism and organ function, which can lead to a decreased tolerance for drug effects.

In this study, the severity of DDIs is considered as the primary research question. The system not only confirms whether an interaction is present; it also confirms whether the interaction is minor, moderate or major. This enhances the clinical usefulness of the system because a major interaction should have a greater impact on the risk score than a minor interaction.

2.1.3 Clinical Decision Support System

A clinical decision support system is a computerized tool designed to help users make safer and better decisions. In the context of polypharmacy, decision-support systems can assist with identifying high-risk prescriptions, detecting drug interactions, and medication safety reviews. note that current polypharmacy decision-support systems vary in design, knowledge source, output and evaluation, and that there is a need for better systems that use multiple approaches [7]. This justifies the proposed system, which uses a DDI knowledge base, machine learning, and an expert-based scoring formula.

2.1.4 Machine Learning

Machine learning was used in this study to assist predictive analysis. Random Forest was chosen as it is suitable for classification and can handle encoded categorical variables. Random Forest is an ensemble learning algorithm that constructs a number of decision trees and merges their predictions to enhance classification accuracy [13]. Machine learning is primarily used in this system for the drug interaction-related model and the adverse drug event prediction model. Other clinical information, including age, liver function, kidney function, gender and diseases, are input by the user.

2.2 Proposed System Approach

The proposed system follows a hybrid approach. It combines knowledge-based interaction lookup, machine learning-supported prediction, and expert-based risk scoring. The proposal originally planned a system where DDI knowledge-base integration, machine learning model development, and EB-PRS scoring work together to calculate individualized polypharmacy risk.

The final implemented workflow is as follows:

1. The user enters patient details, including age, gender, liver function, kidney function, existing diseases, and current medicines.
2. The system validates and cleans the input data.
3. Duplicate drug names are removed to avoid repeated scoring.
4. The drug interaction module checks all possible drug pairs.
5. The system summarizes the number of minor, moderate, and major interactions.
6. The expert-informed risk formula calculates the final risk score.
7. The adverse drug event model predicts possible ADE outcomes using drug, age, gender, and existing disease data.
8. The backend saves the assessment and sends the result to the frontend dashboard.
9. The dashboard displays the risk score, risk level, interaction count, severity summary, ADE predictions, and detailed score breakdown.

This structure allows the system to produce both a numerical score and an interpretable explanation of how the score was generated.

2.2.1 Data Collection and Preprocessing

The system uses two main types of data: drug interaction data and adverse drug event data. The drug interaction data is used to identify interactions between medicine pairs and classify them according to severity. The adverse drug event dataset is used to train the ADE prediction model.

Preprocessing was necessary because raw medical datasets may contain inconsistent names, missing values, duplicate entries, and categorical values that cannot be directly used by machine learning algorithms. The proposal also identifies cleaning, standardization, categorical encoding, and handling incomplete records as important preprocessing steps.

For the drug interaction dataset, drug names were cleaned by trimming spaces and converting names to lowercase. This helped reduce mismatch errors caused by capitalization or formatting differences. Drug pairs were also sorted so that the system would treat Drug A–Drug B and Drug B–Drug A as the same pair. This is important because drug interaction should not depend on the order in which drugs are entered.

For the ADE dataset, missing values were removed from the required fields: Drug, Age, Gender, Existing Diseases, and Predicted_ADE. The categorical variables were then encoded using label encoders. The implementation code confirms that separate encoders were used for drug, disease, age, gender, and target ADE classes.

2.2.2 Drug Interaction Knowledge Base

The drug interaction knowledge base is the foundation of the main risk scoring component. It stores known drug interaction pairs together with severity and description details. During analysis, the system checks every possible pair from the user’s medication list against the knowledge base. If an interaction exists, the interaction is returned with its severity level and description.

The proposal planned the knowledge base as a core part of the system, where drug–drug interaction information is gathered, categorized by severity, and mapped to patient medication profiles for risk assessment. This design is important because drug interactions are not equal in clinical impact. A minor interaction may require monitoring, while a major interaction may require urgent review.

In the implementation, the system identifies interactions using pairwise checking. For example, if the user enters five medicines, the backend checks all possible combinations between those medicines. Each detected interaction is added to the final interaction list, and the severity summary is updated. The severity summary is then passed to the polypharmacy risk score calculation function.

2.2.3 Machine Learning Model for Drug Interaction Prediction

The drug interaction model was developed to support the interaction-related intelligence of the system. The training process used a drug interaction dataset containing Drug_A, Drug_B, and Severity Level. The data was cleaned by removing records with missing values, normalizing drug names, and standardizing severity labels. The implementation created a sorted drug-pair representation so that the same pair would be recognized regardless of order.

A Random Forest classifier was selected for this task because it is suitable for classification problems and performs well with structured data [13]. The model used encoded drug identifiers and pair-based features to learn severity categories. The training script used an 80/20 train-test split and saved the trained model, encoders, and drug list for later system use.

Although the final operational interaction detection depends strongly on the knowledge base for known interactions, the machine learning model supports the system's wider predictive direction. This is consistent with the proposal objective of combining drug interaction data and machine learning techniques to support a dynamic risk scoring framework.

2.2.4 Expert-Based Polypharmacy Risk Score Formula

The main research output of this study is the Expert-Based Polypharmacy Risk Score (EB-PRS). The purpose of this formula is to convert the patient's medication profile and selected clinical risk factors into a single numerical score and an understandable risk level. Since polypharmacy risk is not caused only by the number of medicines, the formula was designed to include medication count, patient age, drug-drug interaction severity, liver function, and kidney function.

Polypharmacy is commonly defined as the concurrent use of five or more medicines, but research shows that medication count alone is not sufficient to describe the actual risk of harm [1], [2]. Drug–drug interactions are one of the major contributors to adverse drug reactions and medication-related complications in patients who take multiple medicines [5]. Older adults are also more vulnerable because ageing, multimorbidity, and reduced organ function can affect how drugs are metabolized and eliminated from the body [1], [6]. Therefore, the formula in this research was designed as a weighted scoring model that gives higher importance to drug–drug interaction severity while still considering other patient-specific factors.

The EB-PRS formula was developed using expert knowledge and verified by Dr. Rohan Jayasooriya. The formula was implemented in the backend risk calculation module of the system. It produces a final score between 0 and 100 and classifies the patient into one of four risk levels: Low, Moderate, High, or Very High.

2.2.4.1 Formula Design Rationale

The formula was designed around five sub-scores. Each sub-score represents one clinically meaningful risk component in polypharmacy assessment. These sub-scores are represented as **S1, S2, S3, S4, and S5**.

Sub-score	Risk factor	Description
S1	Medication count	Measures the risk caused by the number of concurrent medicines
S2	Patient age	Measures age-related vulnerability to medication harm
S3	Drug–drug interaction severity	Measures the cumulative burden of minor, moderate, and major DDIs
S4	Liver function	Measures of risk related to impaired hepatic function

S5	Kidney function	Measures risk related to impaired renal function
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Table 3 Formula Design Rationale

The final score is calculated using weighted values. Drug–drug interaction severity receives the highest weight because the main research purpose is to assess polypharmacy risk based on drug interactions. Medication count and age are also given significant weights because both are strongly associated with polypharmacy-related harm [1], [2]. Liver and kidney function are included because these organs play important roles in drug metabolism and elimination.

$$\text{Risk Score} = \min ((W_1S_1) + (W_2S_2) + (W_3S_3) + (W_4S_4) + (W_5S_5), 100)$$

In the formula you provided for calculating the "Risk Score," the min function plays a key role in limiting the total score to a maximum value of 100. Here's a breakdown of the components:

1. The Formula Structure:

$$\text{Risk Score} = \min ((W_1 \cdot S_1) + (W_2 \cdot S_2) + (W_3 \cdot S_3) + (W_4 \cdot S_4) + (W_5 \cdot S_5), 100)$$

- W_1, W_2, W_3, W_4, W_5 are the weights assigned to different risk factors.
- S_1, S_2, S_3, S_4, S_5 represent the severity scores for each of these risk factors.
- Each pair $(W_n \cdot S_n)$ calculates the weighted score for a specific factor.

2. Weighted Sum:

The sum $(W_1 \cdot S_1) + (W_2 \cdot S_2) + (W_3 \cdot S_3) + (W_4 \cdot S_4) + (W_5 \cdot S_5)$ computes the total risk score by considering the importance (weight) and severity of each factor. The higher the weight and severity of a factor, the greater its contribution to the total score.

3. Capping the Final Score:

The `min` function ensures that the final risk score does not exceed 100, even if the weighted sum exceeds that value.

- For example, if the sum of the weighted scores equals 120, the `min` function will cap it at 100.
- This is useful in cases where multiple high-risk factors (like severe drug interactions) might combine to create a total score higher than 100, which might not be meaningful or useful for analysis.
- By capping the score at 100, it ensures that the risk score always stays within the standard 0–100 range.

Why is this Important?

In practical scenarios, such as evaluating patient risk due to drug interactions, the sum of severity scores can exceed 100 due to multiple severe interactions. By limiting the maximum score to 100, the formula ensures that even in extreme cases, the score is not excessively high, making it easier to interpret and compare across different individuals or situations.

In summary, the `min` function ensures that the final "Risk Score" is bounded within a 0–100 range, preventing runaway scores from extreme risk factors and keeping the result meaningful and easy to understand.

2.2.4.2 Weight Distribution

The weight distribution used in the formula is shown below.

Component	Weight	Justification
Medication count	25%	Polypharmacy risk increases as the number of medicines increases

Patient age	25%	Older adults have higher vulnerability to medication-related harm
DDI severity	30%	Main research focus; severe DDIs directly affect patient safety
Liver function	10%	Impaired liver function may affect drug metabolism
Kidney function	10%	Impaired kidney function may reduce drug elimination

Table 4 Weight Distribution

The total base weight is 100. However, the DDI score is cumulative, meaning that multiple interactions can increase the DDI component. This design was intentionally selected because a patient with several moderate or major interactions should receive a higher risk score than a patient with only one minor interaction.

2.2.4.3 S1: Medication Count Score

The first component, **S1**, measures medication burden. The threshold of five medicines was selected because five or more concurrent medicines is widely used as a common definition of polypharmacy in the literature [1], [2].

Number of medicines	S1 score	Explanation
Less than 5	0.0	Below common polypharmacy threshold
5–7	0.7	Polypharmacy present
8–10	1.0	Higher medication burden
More than 10	1.0 + 0.1 for each additional drug above 10	Very high medication burden

Table 5 Medication Count Score

The implemented logic allows the S1 score to increase beyond 1.0 when the patient takes more than ten medicines. This is because the number of possible drug combinations increases as more medicines are added, which can increase the chance of interactions.

2.2.4.4 S2: Patient Age Score

The second component, **S2**, measures age-related risk. Older patients are more vulnerable to polypharmacy-related problems because ageing can affect drug response, drug clearance, and tolerance to adverse effects [1], [6].

Patient age	S2 score	Explanation
Below 65 years	0.0	Lower age-related risk
65–74 years	0.5	Early geriatric risk category
75–84 years	0.7	Increased vulnerability
85 years and above	1.0	Highest age-related vulnerability

Table 6 S2: Patient Age Score

This scoring method allows the system to increase risk gradually as the patient becomes older. The patient's age is entered directly through the system interface and passed to the backend risk calculation function.

2.2.4.5 S3: Drug–Drug Interaction Severity Score

The third component, **S3**, is the most important part of this research because it represents the drug interaction burden. Existing polypharmacy tools often focus on medication count, but DDI severity provides a more meaningful indication of medication safety risk [5], [8]. For that reason, S3 has the highest formula weight.

The system identifies drug interactions from the entered medication list and summarizes them according to severity. Each interaction severity level is assigned a different score.

Interaction severity	Severity value	Explanation
Major	1.0	Highest risk; may require urgent clinical review
Moderate	0.6	Medium risk; may require monitoring or adjustment
Minor	0.3	Lower risk; usually requires awareness or observation

Table 7 S3: Drug–Drug Interaction Severity Score

The S3 score is calculated as follows:

$$S_3 = (1.0 \times \text{Major Count}) + (0.6 \times \text{Moderate Count}) + (0.3 \times \text{Minor Count})$$

For example, if a patient has one major interaction, two moderate interactions, and one minor interaction, then:

$$S_3 = (1 \times 1.0) + (2 \times 0.6) + (1 \times 0.3) = 2.5$$

This cumulative approach was used because multiple interactions should increase the final risk score. A patient with several interactions should not receive the same score as a patient with only one minor interaction. The final risk score is still capped at 100 to keep the output within a standard range.

2.2.4.6 S4: Liver Function Score

The fourth component, **S4**, measures liver-related medication risk. The liver is responsible for metabolizing many medicines. If liver function is impaired, medicines may remain active in the body for longer periods or may reach toxic levels. Therefore, liver function was included as a patient-specific clinical factor.

Liver function category	S4 score	Explanation
Normal: ALT/AST < 40 IU/L	0.0	No added liver-related risk
Mild risk: ALT/AST 40–80 IU/L	0.3	Mild liver-related risk
Moderate risk: ALT/AST 80–150 IU/L	0.6	Moderate liver-related risk
Severe risk: ALT/AST > 150 IU/L	1.0	Highest liver-related risk

Table 8 S4: Liver Function Score

The liver function value is not predicted by the machine learning model. It is directly entered by the user through the system interface. This design keeps the system practical because liver function is treated as a clinical input rather than an inferred value.

2.2.4.7 S5: Kidney Function Score

The fifth component, **S5**, measures kidney-related medication risk. Kidney function is important because many medicines and drug metabolites are eliminated through the kidneys. Reduced renal function can increase drug accumulation and adverse effects, especially in older adults [6].

Kidney function category	S5 score	Explanation
Stage 1: eGFR 90	0.0	Normal or near-normal kidney function
Stage 2: eGFR 60–89	0.3	Mild reduction
Stage 3a: eGFR 45–59	0.5	Mild to moderate reduction

Stage 3b: eGFR 30–44	0.7	Moderate to severe reduction
Stage 4: eGFR 15–29	0.9	Severe reduction
Stage 5: eGFR below 15	1.0	Highest kidney-related risk

Table 9 S5: Kidney Function Score

Similar to liver function, kidney function is entered directly by the user. The system maps the selected kidney function category into the S5 score and includes it in the final EB-PRS calculation.

2.2.5 Final Risk Level Classification

After calculating the final score, the system maps the value into a risk level. This classification helps users understand the result more easily than a numerical score alone.

Final_risk score	Risk level	Interpretation
0–29	Low	Low polypharmacy risk based on current inputs
30–59	Moderate	Some risk factors are present and should be monitored
60–79	High	Significant medication safety risk is present
80–100	Very High	Serious risk; clinical medication review is strongly recommended

Table 10 Final Risk Level Classification

The risk level is displayed on the dashboard together with the detailed breakdown of S1, S2, S3, S4, and S5. This improves transparency because the user can see how each factor contributed to the final score.

2.2.6 Example Risk Calculation

Assume that a 78-year-old patient takes eight medicines. The system detects one major interaction, two moderate interactions, and one minor interaction. The patient has moderate liver risk and Stage 3a kidney function.

The sub-scores are:

Component	Value
S1: 8 medicines	1.0
S2: age 78	0.7
S3: 1 major, 2 moderate, 1 minor	2.5
S4: moderate liver risk	0.6
S5: Stage 3a kidney function	0.5

Table 11 Risk Calculation

The weighted calculation is:

$$RiskScore = (25 \times 1.0) + (25 \times 0.7) + (30 \times 2.5) + (10 \times 0.6) + (10 \times 0.5)$$

$$RiskScore = 25 + 17.5 + 75 + 6 + 5 = 128.5$$

Since the score is capped at 100:

$$FinalRiskScore = 100$$

Therefore, the risk level is:

$$RiskLevel = VeryHigh$$

This example shows why the capping mechanism is required. Without capping, patients with several interactions may exceed 100. The capped score keeps the final dashboard output standardized while still reflecting the seriousness of cumulative DDI burden.

2.2.7 Importance of the Formula in the System

The EB-PRS formula is important because it connects the research problem directly with the implemented system. It does not treat polypharmacy as only a medication-count problem. Instead, it evaluates the patient's risk using a combination of drug interaction severity and clinical vulnerability factors.

The formula also improves interpretability. Each sub-score has a clear explanation, and each component is shown separately in the dashboard. This is important for clinical decision-support systems because users should be able to understand why a patient received a certain risk level [7]. The system's output is not intended to replace a doctor or pharmacist. Instead, it acts as a decision-support tool that highlights risk and encourages further clinical review.

2.2.8 Summary

The Expert-Based Polypharmacy Risk Score formula is the central contribution of this research. It combines five important factors: medication count, age, drug–drug interaction severity, liver function, and kidney function. Among these, DDI severity has the highest weight because the research focuses mainly on drug interaction-based polypharmacy risk. The final output is a numerical score and a risk level that can be displayed in the system dashboard.

By using this formula, the implemented system provides a more meaningful assessment than simple medicine counting. It supports patient-specific risk interpretation and helps identify patients who may require closer medication review.

2.3 Adverse Drug Event Prediction Model

In addition to calculating the polypharmacy risk score, this research includes an adverse drug event prediction component. This component predicts possible adverse drug events using drug, age, gender, and existing disease information. This was added as an extension to the main research so that the system can show not only the overall risk level but also possible future adverse outcomes.

The ADE model was trained using a Random Forest classifier. The implementation confirms that the dataset fields used were Drug, Age, Gender, Existing Diseases, and Predicted_ADE. Each categorical variable was encoded using LabelEncoder, and the feature matrix was built using encoded drug, age, gender, and disease values. The target variable was the encoded predicted ADE.

The dataset was split into training and testing sets using an 80/20 split. Stratification was applied to preserve the class distribution of the target variable. After testing, the model was retrained on the full dataset for production use and saved as a pickle file along with the encoders. The saved artifacts include the Random Forest model, drug encoder, disease encoder, age encoder, gender encoder, target encoder, metadata file, and evaluation report.

During prediction, the backend checks whether the entered drug and disease are known to the encoders. If a valid match is found, the encoded features are passed into the trained model. The model returns a predicted ADE class and confidence value. This result is displayed in the dashboard as part of the patient's polypharmacy assessment.

2.4 High-Level System Architecture

The system follows a web-based client-server architecture. The front end is developed using Next.js and React, while the backend is developed using Flask. The backend handles analysis requests, drug searching, disease searching, risk score calculation, ADE prediction, assessment retrieval, profile updating, and assessment deletion.

The high-level architecture consists of four main layers:

Layer	Description
User Interface Layer	Collect patient details, medication list, liver function, kidney function, and diseases
API Layer	Receives requests, validates input, and sends responses
Processing Layer	Performs drug interaction lookup, risk score calculation, and ADE prediction
Storage Layer	Stores user profiles and latest polypharmacy assessment records

Table 12 High-Level System Architecture

The proposal's methodology also describes the system in terms of input, processing, and output. Inputs include patient data, medication data, and the DDI knowledge base. Processing includes preprocessing, DDI severity mapping, feature extraction, machine learning model usage, and risk scoring. Outputs include the risk dashboard, alerts, and recommendations.

2.5 Functional and Non-Functional Requirements

2.5.1 Functional Requirements Achieved

The following functional requirements were achieved in the implementation:

Functional Requirement	Implementation Status
User can enter patient age, gender, liver function, and kidney function	Achieved
User can enter multiple medicines	Achieved
System validates minimum and maximum drug count	Achieved
System removes duplicate medicines	Achieved
System searches drug names with suggestions	Achieved
System searches disease names with suggestions	Achieved

System detects drug–drug interactions	Achieved
System summarizes minor, moderate, and major interactions	Achieved
System calculates final risk score	Achieved
System assigns risk level	Achieved
System predicts possible ADEs	Achieved
System saves latest assessment	Achieved
User can view assessment in dashboard	Achieved
User can clear previous assessment	Achieved
Dashboard can export report outputs	Achieved

Table 13 Functional Requirements Achieved

These requirements support the objective of producing a practical decision-support tool rather than only a theoretical model.

2.5.2 Non-Functional Requirements Achieved

The system also considers non-functional requirements. Usability was addressed through a simple web form and dashboard-based result display. Reliability was supported through input validation, exception handling, and cached loading of model files. Performance was improved by loading the drug interaction knowledge base and ADE model once and reusing them during requests. Security was considered by associating assessments with authenticated users and saving user-specific data separately.

The interface was designed for readability because the target users may include elderly patients, caregivers, and healthcare professionals. The dashboard presents score components separately so that the final score can be interpreted clearly. Interpretability is

important in clinical decision-support systems because healthcare users need to trust the output and understand its reasoning [7].

2.6 Social, Security, and Ethical Considerations

The system has a direct social purpose because it supports safer medication management for older adults. Polypharmacy is common among elderly patients and can lead to harmful outcomes if it is not reviewed properly [1]. By identifying risky drug combinations and predicting possible ADEs, the system can help users become more aware of medication safety risks.

From a security perspective, the system handles sensitive patient-related data such as age, gender, medication list, existing diseases, and assessment results. Therefore, access control and safe storage are important. The system stores assessments by user ID, and only the relevant user's latest assessment is retrieved through the API.

From an ethical perspective, the system should not replace medical professionals. The output should be treated as decision-support information, not a final medical diagnosis or prescription decision. Users should be advised to consult a qualified doctor or pharmacist before changing, stopping, or replacing any medicine. This limitation is important because machine learning and rule-based systems can support decision-making, but they cannot fully understand the complete clinical condition of a patient.

2.7 Limitations of the Methodology

The methodology has several limitations. First, the system depends on the quality and completeness of the drug interaction dataset and ADE dataset. If a drug or disease is not present in the dataset, the system may not be able to generate an interaction or ADE prediction for that entry. Second, the ADE model uses encoded categorical features, which means it learns from available categories rather than understanding full clinical context.

Third, the system does not currently process dosage, duration of therapy, route of administration, or laboratory values beyond the selected liver and kidney categories.

Another limitation is that the formula is expert-informed and verified, but wider clinical validation with real patient cases is still required. In future work, the model should be tested with larger clinical datasets and compared with doctor or pharmacist evaluations. The proposal also identifies testing with simulated patient cases and comparison against clinician/pharmacist evaluations as part of the validation direction.

2.8 Tools and Technologies

The implemented system uses the following tools and technologies:

Tool / Technology	Purpose
Python	Data preprocessing, model training, backend logic
Flask	Backend API development
Pandas	Dataset loading and preprocessing
NumPy	Numerical processing
Scikit-learn	Label encoding, train-test split, Random Forest training, evaluation
Pickle	Saving trained models and encoders
Next.js	Frontend application framework
React / TypeScript	User interface implementation
Firestore	User profile and assessment storage
XLSX / jsPDF	Exporting reports from dashboard
CSS / Tailwind-style UI	Styling and responsive design

Table 14 Tools and Technologies

These technologies were selected because they support fast development of an interactive machine learning-based web application. Python and Scikit-learn were suitable for training the machine learning models, while Flask provided a lightweight backend for connecting model logic with the frontend. Next.js and React were suitable for building the patient form, dashboard, and report display components.

2.9 Summary of Methodology

This methodology developed a complete system flow for polypharmacy risk assessment based on drug interaction. The research first identified the clinical need for a DDI-focused risk scoring system. Then, drug interaction data and ADE data were cleaned and prepared. A DDI knowledge base was integrated to detect interaction severity, and a Random Forest model was used for ADE prediction. The final risk score was calculated using an expert-informed formula that combines medication count, patient age, DDI severity, liver function, and kidney function.

The outcome of this methodology is a web-based clinical decision-support style system that provides a risk score, risk level, interaction summary, and predicted adverse drug event output. This directly supports the main research objective of creating a patient-specific and interpretable polypharmacy risk score system based on drug interaction.

3.RESULTS AND DISCUSSION

This chapter presents the results and discussion of the implemented Polypharmacy Risk Score System Based on Drug Interaction. The results are discussed based on the final working system, backend implementation, machine learning model outputs, expert-based risk calculation, adverse drug event prediction, and user-facing dashboard. The main aim of the system was to calculate a patient-specific polypharmacy risk score by giving priority to drug–drug interaction severity while also considering patient age, medication count, liver function, and kidney function.

The system was developed to address the limitation found in many existing polypharmacy assessment approaches, where risk is often estimated mainly through drug count, age, or general comorbidity information. As discussed in the literature, polypharmacy is strongly associated with drug interactions, hospitalization, medication-related harm, and other adverse outcomes among older adults [1]. Therefore, the implemented system was

designed to provide a more specific risk assessment by combining drug interaction severity with patient-specific clinical factors.

The final system produces four main outputs. First, it identifies drug–drug interactions from the patient’s medication list. Second, it summarizes the severity of detected interactions. Third, it calculates the final polypharmacy risk score and assigns a risk level. Fourth, it predicts possible adverse drug events using the patient’s drug, age, gender, and existing disease information. These outputs are displayed through a dashboard so that the result can be interpreted easily by patients, caregivers, and healthcare professionals.

3.1 Results

3.1.1 System Implementation Results

The final system was implemented as a web-based clinical decision-support style application. The frontend allows the user to enter patient information, medication details, liver function, kidney function, and existing diseases. The backend performs input validation, drug interaction checking, risk score calculation, ADE prediction, and assessment saving. This structure matches the requirement of a decision-support system because it accepts patient-specific data, processes clinical and medication information, and returns an interpretable output [7].

The implemented system successfully achieved the following core functions:

Function	Result
Patient profile input	Achieved
Medication list input	Achieved
Drug name suggestion/search	Achieved
Disease name suggestion/search	Achieved

Duplicate drug removal	Achieved
Minimum and maximum drug validation	Achieved
Drug–drug interaction detection	Achieved
Severity summary generation	Achieved
Expert-based risk score calculation	Achieved
Risk level classification	Achieved
ADE prediction	Achieved
Dashboard result display	Achieved
Assessment saving and retrieval	Achieved
Report export support	Achieved

Table 15 System Implementation Results

The result confirms that the final product is not limited to a theoretical risk model. It works as a complete application where the user can submit clinical and medication details and receive an assessment result. This is important because polypharmacy management requires practical tools that can support clinical review rather than only research-level scoring concepts [7].

3.1.2 Drug Interaction Detection Results

The drug interaction detection component is the central part of the research. When the user enters multiple medicines, the system checks possible drug pairs and identifies whether known interactions exist. Each detected interaction is returned with the drug pair, severity level, and available description. The interaction severity is then summarized into categories such as minor, moderate, and major.

This result directly supports the research objective because the system does not treat all interactions equally. A major interaction contributes more to the final risk score than a moderate or minor interaction. This is important because literature shows that drug–drug

interactions are a major contributor to medication-related harm in polypharmacy [5], [8]. A simple interaction-present or interaction-absent approach is not enough because the clinical seriousness of interactions can differ significantly between patients.

The implemented severity-based approach improves the meaning of the final result. For example, a patient with several minor interactions may receive a lower DDI contribution than a patient with fewer but more severe interactions. This allows the system to reflect the actual seriousness of the medication combination more accurately.

Severity Level	Interpretation in the System	Clinical Meaning
Minor	Low contribution to DDI risk score	Usually requires awareness or basic monitoring
Moderate	Medium contribution to DDI risk score	Requires closer monitoring or clinical review
Major	High contribution to DDI risk score	Requires urgent medication review or professional attention

Table 16 Drug Interaction Severity Interpretation

This result supports the research gap identified in Chapter 1, where existing systems were found to be limited because they often do not integrate DDI severity into a patient-specific risk score. The result also supports the view that decision-support systems should provide meaningful clinical information rather than only general alerts [7], [9].

3.1.3 Polypharmacy Risk Score Calculation Results

The final polypharmacy risk score is calculated using five sub-score components: medication count, patient age, DDI severity, liver function, and kidney function. The

formula uses a weighted structure where DDI severity receives the highest weight because the main research purpose is to assess polypharmacy risk based on drug interaction.

The implemented scoring model uses the following component weights:

Component	Factor	Weight
S1	Medication Count	25
S2	Patient Age	25
S3	Drug–Drug Interaction Severity	30
S4	Liver Function	10
S5	Kidney Function	10
Total		100

Table 17 Polypharmacy Risk Score Calculation Results

The final risk score is calculated on a 0–100 scale. This makes the output easy to understand because the score can be directly mapped into a risk category. The risk levels used in the system are shown below.

Table 3.2: Risk Score Classification

Risk Score Range	Risk Level	Interpretation
0–29	Low	Minimal clinical concern
30–59	Moderate	Requires monitoring
60–79	High	Significant risk
80–100	Very High	Critical condition requiring intervention

Table 18 Risk Score Classification

The main result of this scoring method is interpretability. The system does not only display the final score; it also shows the contribution of each component. This makes the result more transparent and suitable for clinical decision-support use. Interpretability is important because healthcare-related systems should help users understand why a particular risk level was assigned [7].

The score also reflects the clinical importance of medication burden. Polypharmacy is commonly defined using five or more medicines [1], [2]. However, this system improves on simple medication counting by combining the number of drugs with DDI severity and organ function. This is more suitable for elderly patients because reduced liver and kidney function can increase medication-related risks [6].

3.1.4 Risk Score Demonstration

The following scenarios demonstrate the behavior of the implemented scoring formula. These are not clinical trial results; they are controlled functional demonstrations created to verify that the formula responds correctly to changes in medication count, age, interaction severity, liver function, and kidney function.

Scenario	Patient profile	S1	S2	S3	S4	S5	Score	Risk level
A	55 years, 3 drugs, no DDI, normal liver, stage 1 kidney	0.0	0.0	0.0	0.0	0.0	0.0	Low
B	70 years, 5 drugs, 1 moderate DDI, mild liver,	0.7	0.5	0.6	0.3	0.3	54.0	Moderate

	stage 2 kidney							
C	80 years, 8 drugs, 1 major DDI, normal liver, stage 3a kidney	1.0	0.7	1.0	0.0	0.5	72.5	High
D	86 years, 10 drugs, 2 major + 1 moderate DDI, severe liver, stage 4 kidney	1.0	1.0	2.6	1.0	0.9	100.0	Very High

Table 19 Risk Score Demonstration

3.1.5 Adverse Drug Event Prediction Results

The system includes an additional adverse drug event prediction component. This model predicts possible ADE outcomes based on the selected drug, patient age, gender, and existing diseases. Random Forest was used as the machine learning algorithm because it is suitable for structured classification problems and can handle multiple encoded categorical inputs [13].

The ADE model used the following input features:

Feature	Description
Drug	Medicine selected by the user
Age	Patient age category/value from the dataset
Gender	Patient gender
Existing Diseases	Patient's existing disease condition

Table 20 Adverse Drug Event Prediction Results

The target output of the model is:

Target	Description
Predicted_ADE	Possible adverse drug event class

The training process used label encoding for categorical variables and an 80/20 train-test split. The model was trained using a Random Forest classifier with 200 estimators, maximum depth of 20, and minimum samples split of 5. After evaluation, the model artifacts and encoders were saved for production use. This supports the machine learning requirement of the research and allows the backend to load the trained model during prediction.

The ADE prediction result adds value to the main polypharmacy risk score because it gives the user more specific awareness of possible future adverse outcomes. Existing polypharmacy systems are often descriptive, while this system includes a predictive extension. This aligns with recent research interest in using machine learning to validate medication safety features and identify patients at higher risk of medication-related harm [11], [14].

Output Field	Description
--------------	-------------

Drug	Drug used for ADE prediction
Existing Disease	Disease condition matched with the drug
Predicted ADE	Possible adverse drug event predicted by the model
Confidence	Model confidence percentage for the predicted event

Table 21 ADE Prediction Output Format

The ADE component is not intended to replace clinical diagnosis. Instead, it works as a supportive prediction that can help users identify possible adverse outcomes early. A doctor or pharmacist should verify the result before any medication change is made.

3.1.6 Dashboard Result Output

The dashboard was developed to present the final results clearly. It displays the patient summary, medication list, number of detected interactions, interaction severity summary, overall risk score, risk level, risk score breakdown, and ADE predictions.

The dashboard output includes the following result sections:

Dashboard Section	Displayed Information
Patient Information	Age, gender, liver function, kidney function, diseases
Medication List	Entered drugs after validation
Interaction Summary	Number of detected interactions
Severity Summary	Minor, moderate, and major interaction counts
Risk Score	Final calculated score
Risk Level	Low, Moderate, High, or Very High
Score Breakdown	S1, S2, S3, S4, and S5 contribution
ADE Predictions	Predicted adverse drug events and confidence
Export Option	PDF/Excel-style report output

Table 22 Dashboard Result Output

The dashboard improves usability because users do not need to interpret raw model output manually. Instead, they receive a structured summary. This is important for clinical decision-support systems because the output must be understandable and actionable [7]. The dashboard also supports communication between patients, caregivers, and healthcare professionals by presenting the assessment in a report-style format.

3.1.7 Testing Results

Testing was performed to confirm that the main system functions work as expected. The testing process focused on input validation, interaction detection, risk calculation, ADE prediction, dashboard rendering, and assessment storage.

Test Case	Expected Result	Actual Result	Status
Submit without user ID	Error message should be returned	Error returned	Pass
Submit fewer than two distinct drugs	Error message should be returned	Error returned	Pass
Submit more than maximum allowed drugs	Error message should be returned	Error returned	Pass
Submit duplicate drug names	Duplicates should be removed	Duplicates removed	Pass
Submit valid drug list	Interactions should be checked	Interactions checked	Pass
Submit age below valid range	Validation should prevent processing	Error returned	Pass

Submit valid liver/kidney function	Scores should be calculated	Scores calculated	Pass
Submit existing diseases	ADE prediction should be attempted	ADE predictions generated for known entries	Pass
View dashboard after analysis	Latest assessment should display	Assessment displayed	Pass
Clear assessment	Stored assessment should be removed	Assessment cleared	Pass

Table 23 Functional Testing Summary

The testing results show that the implemented system performs the expected workflow correctly. Input validation is especially important because incorrect patient or medication data can affect the final score. In healthcare-related systems, safe handling of invalid inputs is necessary to reduce misleading outputs and improve user trust [7].

3.2 Discussion

The results show that the system successfully addresses the main research gap by integrating drug–drug interaction severity into a patient-specific polypharmacy risk score. Unlike simple medication-count systems, the implemented model considers how dangerous the medication combinations are. This is an important improvement because polypharmacy risk does not depend only on the number of medicines but also on the type and severity of interactions between them [1], [5].

The use of an expert-based formula also improves the interpretability of the system. A purely machine learning-based model may produce predictions without a clear explanation, which can be difficult to trust in healthcare contexts. In contrast, the proposed system shows each component of the score. This makes it easier for healthcare professionals to understand the basis of the result. It also helps patients and caregivers understand why the system identified a certain risk level.

The ADE prediction component further strengthens the system. The main risk score identifies the overall level of polypharmacy risk, while the ADE model predicts possible adverse event outcomes. This makes the system more useful than a descriptive risk calculator. Recent studies show that machine learning can be useful for validating medication safety features and identifying high-risk medication patterns [11], [14]. Therefore, the inclusion of ADE prediction is relevant to the future direction of clinical decision-support systems.

The final system also supports practical usability. The frontend form allows structured data entry, and the dashboard provides clear outputs. This is consistent with the need for decision-support tools that are understandable and usable in real clinical workflows [7]. The system can support early identification of high-risk patients and can help users decide when professional medication review is necessary.

However, the system has limitations. First, the quality of the result depends on the completeness of the drug interaction and ADE datasets. If a drug or disease is not available in the dataset, the system may not generate a prediction for that entry. Second, the ADE model uses encoded categorical values and does not yet consider dosage, duration, route of administration, laboratory values, or full medical history. Third, the risk scoring formula was expert-informed and verified, but wider clinical validation is still required using real patient cases.

Another limitation is that the system should not be treated as a replacement for doctors or pharmacists. Medication decisions require full clinical judgment. The system should be used as a decision-support tool that highlights possible risks and encourages professional review. This is consistent with general recommendations that medication review and deprescribing decisions should be made carefully by considering the patient's full clinical condition, treatment goals, and risk-benefit balance [6].

3.3 Research Findings

The main findings of the research are as follows:

1. The system successfully calculates a patient-specific polypharmacy risk score using drug count, age, DDI severity, liver function, and kidney function.
2. Drug–drug interaction severity improves the meaningfulness of the final score because the system can distinguish between minor, moderate, and major interaction burden.
3. The expert-based formula provides a transparent scoring structure that is easier to interpret than a black-box prediction model.
4. The ADE prediction model adds predictive value by identifying possible future adverse drug events based on drug, age, gender, and existing disease.
5. The dashboard improves usability by presenting risk score, risk level, interaction summary, score breakdown, and ADE predictions in one place.
6. The system supports the main research objective by providing a dynamic, implementation-based, and clinically interpretable polypharmacy risk assessment framework.

3.4 Summary

This chapter discussed the results of the implemented Polypharmacy Risk Score System Based on Drug Interaction. The results show that the system achieved its main objective by detecting drug interactions, summarizing severity, calculating a weighted risk score, assigning a risk level, and predicting possible adverse drug events. The system also provides a dashboard that presents the results in an understandable and clinically useful manner.

The discussion shows that the system contributes to the research gap by moving beyond simple medication counting and integrating DDI severity into a personalized risk score. It also includes a machine learning-based ADE prediction feature, making the system more predictive and practical. Although further clinical validation is required, the implemented system provides a strong foundation for safer medication review and polypharmacy risk awareness among geriatric patients.

4. CONCLUSION

This research successfully designed and implemented a Polypharmacy Risk Score System Based on Drug Interaction as part of the wider Smart Polypharmacy and Vitamin Recommendation System for Geriatric Patients. The main purpose of this study was to calculate a patient-specific polypharmacy risk score by considering not only the number of medicines, but also the severity of drug–drug interactions, patient age, liver function, and kidney function. This directly addresses the research problem because patients with the same number of medicines may have different risk levels depending on the seriousness of their drug interactions and clinical condition.

The developed system follows a hybrid approach by combining a drug interaction knowledge base, machine learning-supported prediction, and an expert-informed risk scoring formula. The system accepts patient details, current medications, liver function, kidney function, gender, and existing diseases through a user-friendly web interface. It then validates the inputs, removes duplicate medicines, checks possible drug pairs, summarizes interaction severity, calculates the final polypharmacy risk score, assigns a risk level, and predicts possible adverse drug events. These outputs are displayed through a dashboard that presents the risk score, risk level, interaction summary, ADE predictions, and score breakdown in an understandable manner. This supports the project objective of creating an interpretable clinical decision-support style system for medication safety awareness.

The major contribution of this study is the integration of drug–drug interaction severity into the final risk score. Many existing approaches mainly depend on medication count, age, or general clinical information, but this research gives higher importance to interaction severity because severe interactions can increase the likelihood of medication-related harm. By assigning different values to minor, moderate, and major interactions, the system provides a more meaningful assessment than a simple medicine-counting method. The use of an expert-informed formula also improves transparency, since users can understand how each factor contributes to the final risk level.

The implementation results show that the system achieved its main functional requirements. It can collect patient and medication data, detect drug–drug interactions, summarize severity levels, calculate a weighted risk score, classify patients into low, moderate, high, or very high-risk categories, predict possible adverse drug events, save assessments, and display results in a dashboard. Therefore, the final outcome is not only a theoretical model, but also a working full-stack application that can support medication review and risk awareness among geriatric patients.

However, studying has several limitations. The accuracy and usefulness of the system depend on the completeness and quality of the drug interaction and ADE datasets. If a drug, disease, or interaction is not available in the dataset, the system may not generate a complete prediction. The ADE model also uses encoded categorical values and does not yet consider dosage, duration of therapy, route of administration, full laboratory results, or complete medical history. In addition, although the risk scoring formula was expert-informed and verified, wider clinical validation using real patient cases is required before applying the system in a real healthcare environment. The system should therefore be used only as a decision-support tool and should not replace the judgement of doctors, pharmacists, or other qualified healthcare professionals.

Future improvements can strengthen the system further by expanding the drug interaction knowledge base, adding more ADE data, including dosage and treatment duration, integrating additional laboratory values, and validating the system with real clinical cases. The system could also be improved through integration with electronic health records, mobile application support, multilingual interfaces, and more explainable machine learning outputs. These improvements would make the system more reliable, practical, and suitable for use in medication safety screening.

In conclusion, this research provides a practical and interpretable solution for assessing polypharmacy risk in geriatric patients. By combining drug interaction severity, patient-specific clinical factors, expert-based scoring, machine learning-supported ADE prediction, and dashboard-based reporting, the system contributes to safer medication

management. Although further validation and enhancement are required, the developed system provides a strong foundation for future clinical decision-support tools focused on reducing medication-related risks in elderly patients.

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6.APPENDICES

Appendix A: Key Backend Functions

This appendix summarises selected implementation logic from the polypharmacy risk model. The actual implementation includes additional error handling, caching, Firestore calls, and model loading.

```
def calculate_polypharmacy_risk(drug_count, age, severity_summary, liver_function,
kidney_function):
    s1 = calculate_s1_score(drug_count)
    s2 = calculate_s2_score(age)
    s3, total_ddi_count = calculate_s3_score(severity_summary)
    s4 = calculate_s4_score(liver_function)
    s5 = calculate_s5_score(kidney_function)
    risk_score = min((25*s1) + (25*s2) + (30*s3) + (10*s4) + (10*s5), 100.0)
    if risk_score < 30:
        risk_level = "Low"
    elif risk_score < 60:
        risk_level = "Moderate"
    elif risk_score < 80:
        risk_level = "High"
    else:
        risk_level = "Very High"
    return risk_score, risk_level
```

Appendix B: API Request Structure

```
POST /api/polypharmacy/analyze
{
  "userId": "authenticated-user-id",
  "drugs": ["Aspirin", "Warfarin", "Metformin"],
  "age": 70,
  "gender": "male",
  "liverFunction": "Mild risk (40-80 IU/L)",
  "kidneyFunction": "Stage 2: eGFR of 60-89",
  "existingDiseases": ["Diabetes", "Hypertension"]
}
```

Appendix C: API Response Structure

```
{
  "assessmentId": "...",
  "drugCount": 5,
  "interactionsFound": 2,
  "interactions": [{"drugA": "...", "drugB": "...", "severity": "Major"}],
  "severitySummary": {"Minor": 0, "Moderate": 1, "Major": 1},
  "adePredictions": [{"drug": "...", "disease": "...", "predictedADE": "...",
"confidence": 65.2}],
  "riskCalculation": {
    "scores": {"s1": 0.7, "s2": 0.5, "s3": 1.6, "s4": 0.3, "s5": 0.3},
    "riskScore": 84.0,
    "riskLevel": "Very High"
  }
}
```

Appendix D: Frontend Assessment Types

The frontend defines TypeScript structures for the assessment response. This reduces runtime mistakes because the dashboard can rely on typed fields such as interactions, severitySummary, adePredictions, and riskCalculation.

```
type Interaction = {
  drugA: string;
  drugB: string;
  severity: string;
  description?: string;
  ddinterIdA?: string;
  ddinterIdB?: string;
};

type RiskCalculation = {
  scores: { s1: number; s2: number; s3: number; s4: number; s5: number };
  weights: { drugWeight: number; ageWeight: number; ddiWeight: number; liverWeight:
number; kidneyWeight: number };
  drugCount: number;
  ddiCount: number;
  riskScore: number;
  riskLevel: string;
};
```

Appendix E: Controller Validation Logic

The controller rejects incomplete requests before analysis. This protects the model layer from invalid input and gives the frontend a clear error response.

```
if not user_id:
    return jsonify({"message": "userId is required"}), 400
if not isinstance(drugs, list):
    return jsonify({"message": "drugs must be a list"}), 400
if not age or not isinstance(age, int) or age < 0:
    return jsonify({"message": "age is required and must be a positive integer"}), 400
if len(sanitized_drugs) < MIN_DRUGS:
    return jsonify({"message": "Provide at least 2 distinct drugs"}), 400
if len(sanitized_drugs) > MAX_DRUGS:
    return jsonify({"message": "Maximum of 20 drugs is allowed"}), 400
```

Appendix F: Report Export Content

The dashboard export includes multiple sections so that the assessment can be reviewed outside the application. The Excel workbook separates patient summary, medications, drug interactions, severity summary, ADE predictions, and risk analysis. The PDF report provides a compact clinical summary with the final risk badge and a disclaimer.

Export section	Content included
Patient Summary	Report reference, report date, patient name, age, gender, organ function, existing diseases
Medication List	Numbered list of all analysed medicines
Drug Interactions	Drug A, Drug B, severity, and description
Severity Summary	Major, Moderate, and Minor interaction counts
ADE Predictions	Drug, disease, predicted ADE, and confidence
Risk Analysis	S1-S5 sub-scores, weights, weighted components, risk score, and level

Table 24 Report export content

Appendix G: Extended Test Case Matrix

The following extended matrix can be used during future implementation review to confirm that each major function behaves as intended across normal and abnormal inputs.

Test ID	Input condition	Expected output
TC-01	Empty drug search query	Return empty item list
TC-02	Valid partial drug name	Return matching drug suggestions
TC-03	Duplicate medicines entered	Remove duplicates before analysis
TC-04	Only one medicine entered	Return validation error
TC-05	Twenty-one medicines entered	Return maximum drug validation error
TC-06	Severe liver selected	Automatically include Liver Issues in disease list
TC-07	Stage 5 kidney selected	Automatically include Kidney Issues in disease list

TC-08	No interactions found	Return zero severity counts and S3 = 0.0
TC-09	Major interaction found	Increase Major count and S3 by 1.0 per major interaction
TC-10	Unknown ADE drug	Skip ADE prediction for unsupported drug
TC-11	Existing assessment available	Update existing Firestore document
TC-12	No existing assessment	Create new Firestore document
TC-13	Dashboard PDF export clicked	Generate PDF report with patient and risk details
TC-14	Dashboard Excel export clicked	Generate workbook with multiple sheets

Table 25 Extended test case matrix

Appendix H: Future Enhancement Plan

- Validate the risk score using anonymised clinical medication records and pharmacist-reviewed outcomes.
- Add local medicine brand-name mapping to improve Sri Lankan usability.
- Introduce historical assessment tracking to show whether risk increases or decreases after prescription changes.
- Add Sinhala and Tamil language support for patient-facing explanations.
- Improve ADE model evaluation by reporting accuracy, precision, recall, F1-score, and confusion matrix on a held-out dataset.
- Implement role-based access for patients, caregivers, doctors, and administrators.
- Integrate with hospital prescribing systems or pharmacy dispensing records when ethical and regulatory approvals are available.

Appendix I: Deployment and Validation Checklist

Before the proposed system is used outside the academic demonstration environment, the following deployment and validation checklist should be completed. These items are included to show awareness that healthcare-related software requires stronger controls than ordinary information systems.

Area	Checklist item	Reason
Clinical validation	Compare system risk categories with pharmacist-reviewed medication profiles	Confirms whether the score agrees with expert judgement
Model evaluation	Evaluate ADE model with independent test data	Reports actual predictive performance before deployment
Data quality	Expand medicine dictionary with generic names, brand names, and local spelling variants	Improves matching for Sri Lankan users
Security	Apply authentication, authorisation, HTTPS, and database rules	Protects sensitive patient information
Auditability	Log assessment creation, updates, export, and deletion	Supports accountability and troubleshooting
Usability	Test form and dashboard with caregivers, pharmacists, and doctors	Ensures the interface is understandable
Safety	Add clear disclaimers and escalation guidance for high-risk results	Prevents unsafe self-medication decisions
Maintenance	Schedule periodic knowledge-base and model updates	Keeps interaction data and prediction logic current

Table 26 Deployment and validation checklist

The checklist also supports future research evaluation. In a clinical validation study, the system output should be compared with pharmacist medication review, hospitalisation history, and documented adverse drug events. The comparison should not only evaluate score accuracy but also measure whether the dashboard helps users understand why a medication profile is risky.

A safe deployment plan should begin with a closed pilot using anonymised or consented data. After technical and clinical review, the system can be tested in a supervised setting where clinicians review the generated reports before any recommendation is communicated to patients. This staged deployment protects patients while allowing the system to be improved using real-world feedback.

The final long-term goal is not to automate prescribing decisions, but to provide a reliable screening layer that supports better conversations among doctors, pharmacists, caregivers, and elderly patients. When used responsibly, the system can help identify patients who need medication review sooner and can reduce the chance that serious interactions remain unnoticed.